FOR A SMOOTH MORPHISM THE CONORMAL SEQUENCE OF A COMMUTATIVE TRIANGLE OF MORPHISMS IS SHORT EXACT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Statement span 7406-7414 and proof span 7416-7426 come from the halfA pool card operations/pool-cards/stacks-morphisms-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The proof consists of the algebraic reformulation and its citation, as the author wrote it; no step was added or reordered.

Editorial note. 3 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- The conormal sequence of relative differentials for a commutative triangle of morphisms, which is exact at the middle and right terms, cited from the same chapter. Cited at proof lines 7408-7414. Supplies the sequence whose exactness on the left has to be proved.
- The algebraic statement that for ring maps with the second map smooth the corresponding sequence of modules of differentials is exact, cited from the algebra chapter. Cited at proof lines 7416-7426. Gives the affine computation from which the geometric statement follows.

Source reference. Original lemma 00RS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1014–1014. Same-chapter reference used in the retained author text at source line(s) 6645, 7423. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 02K4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3359–3359. Same-chapter reference used in the retained author text at source line(s) 7425. It is supplied only as a navigation pointer

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Source reference. Original lemma 01UX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2426–2426. Same-chapter reference used in the retained author text at source line(s) 6688, 7413, 7994. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let $f : X \to Y$, $g : Y \to S$ be morphisms of schemes. Assume f is smooth. Then*

$$0 \to f^*\Omega_{Y/S} \to \Omega_{X/S} \to \Omega_{X/Y} \to 0$$

(see Lemma 01UX) is short exact.

Proof. The algebraic version of this lemma is the following: Given ring maps $A \to B \to C$ with $B \to C$ smooth, then the sequence

$$0 \to C \otimes_B \Omega_{B/A} \to \Omega_{C/A} \to \Omega_{C/B} \to 0$$

of Algebra, Lemma 00RS is exact. This is Algebra, Lemma 02K4. □